Can large language models process linguistic deference?

Although language models' performance on linguistic dependencies has been widely studied [1,2,3], honorific dependencies remain underexplored. To address this gap, this study investigates how large language models process honorific dependency in Korean. Honorification refers to the linguistic marking of deference toward a referent or addressee. When the subject is an honorable referent such as *sensayngnim* 'teacher', the nominative case appears as *-kkeyse* rather than plain *-i/-ka*, and honorific suffix *-si* is added to the verb stem, (1a). Previous research has shown that sentences lacking these honorific markings, even with an honorific-referent subject (1c), are also acceptable to native speakers [3,4]. However, it is considered unacceptable if *-kkeyse* is present without *-si* on the verb, (1b) vs. (1d) [5]. To test whether language models capture this asymmetry, we evaluate both encoder- and decoder-based models on honorific dependency patterns. Our results show that models broadly reflect human judgments: SUBJECT-ONLY HONORIFIC sentences (1d) are treated as least acceptable, while sentences with full marking (ALL HONORIFIC, 1a), no marking (NO HONORIFIC, 1c), or VERB-ONLY HONORIFIC marking (1b) are judged more acceptable.

Methods. We constructed experiment items varying in honorific markings (ALL, VERB-ONLY, NONE, SUBJ-ONLY) and subject-verb distance (CLOSE, FAR), resulting in 800 test sentences. In the FAR condition, 3-word adverbial phrases were inserted between the subject and the verb phrase to examine distance effects on model performance. We selected 7 language models: 2 Korean-focused encoder-based models (KoBERT, KoELECTRA), 3 Korean-focused decoder-based models (SOLAR, Polyglot-Ko, KoGPT3), and 2 multilingual decoder-based models (Gemma3, Llama3). Accuracy was calculated by assigning a score of 1 per item if the model correctly predicted minimum verb probability for the SUBJ-ONLY HONORIFIC condition. Surprisal was calculated on the verb, the locus of honorific morpheme *-si*, with which *-kkeyse* on the subject must co-occur. For encoder-based models, each token in the verb was masked individually, and its negative log probability was computed from the masked prediction. For decoder-based models, surprisal was computed as the negative log probability of each token given all preceding context. For both model types, the probabilities were averaged across the tokens in the verb.

Results. Models showed varied performance, with overall accuracy ranging from 53% (KoBERT) to 98% (Polyglot-Ko). Across distance conditions, accuracy was generally lower at FAR than CLOSE distance, except for Polyglot-Ko, which maintained consistent performance across both distance conditions (Figure 1). Models showed substantially higher surprisal when honorific marking occurred only in the subject (i.e., SUBJ-ONLY HONORIFIC condition) (Figure 2; $\beta = 1.93$, $p < .001$). Pairwise comparisons using estimated marginal means showed that all models except SOLAR significantly distinguished the SUBJ-ONLY condition from all other conditions. A marginal effect of distance was also observed, with surprisal slightly lower at FAR than CLOSE distance condition ($\beta = -0.46$, $p = .078$). Significant three-way interactions between the honorific condition, distance, and model type indicated that distance effects varied substantially by both condition types and models, with Gemma3 showing the largest variation (SD=3.19).

Discussion. All language models except SOLAR processed honorific dependencies in a largely human-like manner. The models judged SUBJ-ONLY HONORIFIC sentences to be less acceptable than the other conditions, aligning with native-speaker intuitions. The finding that CLOSE sentences are more surprising than FAR sentences may seem counterintuitive since longer dependencies are typically assumed to increase uncertainty. However, given that verb-phrase adverbials in Korean generally occur right before the verb phrase they modify, models may have interpreted FAR conditions to be more natural, resulting in lower surprisal. Lastly, SOLAR's poor performance is likely due to its narrow training domain. Unlike other models trained on diverse genres and registers, SOLAR was trained primarily on question-answer data. This suggests that the nature and diversity of the training data strongly influence a model's ability to capture human-like linguistic patterns.

Table 1. Example experimental materials e.g., 'The teacher went home.'

Condition	Sentence			Grammatical
(1a) ALL HONORIFIC	Sensayngnim-**kkeyse** teacher-NOM.HON	cip-ey house-to	ka-**si**-ess-ta. go-HON-PST-DECL	✅
(1b) VERB-ONLY HONORIFIC	Sensayngnim-**i** teacher-NOM	cip-ey house-to	ka-**si**-ess-ta. go-HON-PST-DECL	✅
(1c) NO HONORIFIC	Sensayngnim-**i** teacher-NOM	cip-ey house-to	ka-ss-ta. go-PST-DECL	✅
(1d) SUBJ-ONLY HONORIFIC	Sensayngnim-**kkeyse** teacher-NOM.HON	cip-ey house-to	ka-ss-ta. go-PST-DECL	❌

DECL = declarative sentential ending, HON = honorific verbal suffix, NOM = nominative case, NOM.HON = honorific nominative case, PST = past tense

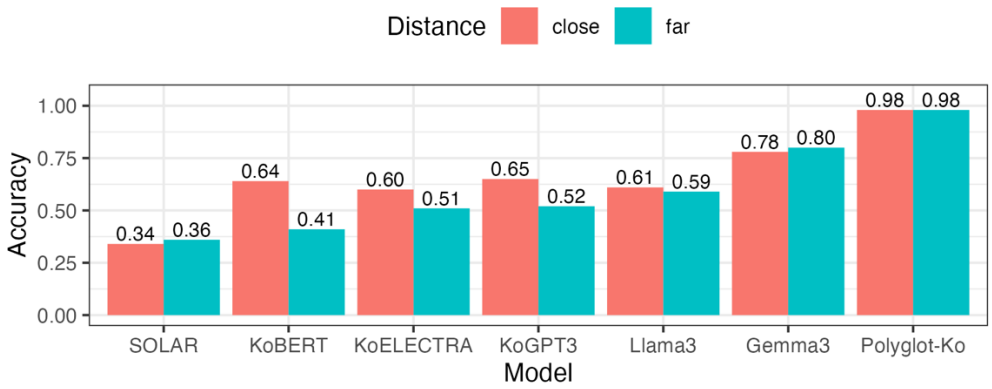

Figure 1. Model accuracy in identifying ungrammatical SUBJ-ONLY honorifics across distance conditions.

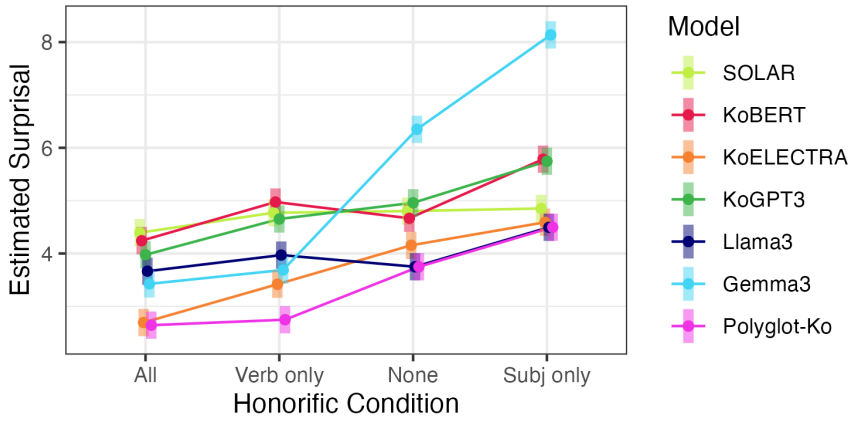

Figure 2. Estimated surprisal of the verb with 95% CIs across honorific conditions.

Selected references. [1] Linzen, T., Dupoux, E., & Goldberg, Y. (2016). Assessing the ability of LSTMs to learn syntax-sensitive dependencies. [2] Wilcox, E., Levy, R., Morita, T., & Futrell, R. (2018). What do RNN language models learn about filler–gap dependencies? [3] Song, S., Choe, J., & Oh, E. (2019). An empirical study of honorific mismatches in Korean. [4] Noh, K., Song, S., & Oh, E. (2024). How language models understand honorific mismatches in Korean. [5] Yoon, J. H-S. (2005). Non-morphological determination of nominal particle ordering in Korean.